

HOURHOUSE

Adult ESL · Newark, NJ

LESSON PLAN

The Vitruvian Approximation

From Ancient Egypt to Leonardo da Vinci

Source Text	The Vitruvian Approximation: Rational Circle-to-Square Transitions in the dm^3 Framework — Pablo Nogueira Grossi 2026 https://doi.org/10.5281/zenodo.19984522
Video	The Beautiful Story Behind Da Vinci's Vitruvian Man (16 min 29 sec) — youtube.com/watch?v=MHd8C8X0BIM
Level	Upper-Intermediate to Advanced (B2–C1)
Skills	Listening · Reading · Speaking · Writing · Vocabulary · Critical Thinking
Duration	2 hours (120 minutes)
Schedule	2 hrs/day · Monday–Thursday · 8 hrs/week
Topic	Mathematics · History of Science · Art & Philosophy · Academic English
Lesson Objective	Students will be able to discuss the history of squaring the circle, read and respond to an academic text written by their teacher, and express mathematical and philosophical ideas in spoken and written English.
Enabling Skills	Vocabulary: approximation, rational, transcendental, convergent, proportion, harmony, operator, conjecture, framework, formalize Grammar: passive voice (was drawn, is defined as, has been solved); academic hedging (it remains an open problem; the answer is: partly); present perfect for historical reference Pronunciation: stress in academic multi-syllable words — ap·PROX·i·MA·tion, tran·SCEN·den·tal, con·VER·gent
Materials	QR code (below) for video access Vocabulary worksheet · Discussion cards



Scan to watch the video

The Beautiful Story Behind Da Vinci's Vitruvian Man — 16 min 29 sec youtube.com/watch?v=MHd8C8X0BIM

Activity Plan

STAGE 1 · Warm-Up / Review (20 minutes)

Activity: Video Viewing — The Vitruvian Man Documentary

Before watching (3 min): Teacher writes these questions on the board. Students read silently before the video starts.

1. Who was Vitruvius? What did he believe about the human body?
2. What is the connection between a circle and a square in Da Vinci's famous drawing?
3. What ancient problem was Leonardo trying to solve?
4. What does the word proportion mean to you?

Watch first 8 minutes of the video. Students take brief notes.

After watching — Pair Discussion (5 min): Students compare answers, then share with the class. Teacher draws out: circle = divine / infinite / perfect; square = stability / order / human; the drawing tries to bridge both. This connects directly to the paper's core question — can we rationally approximate the transition from circle to square?

Teacher bridge (2 min): “The man who wrote the paper we are reading today — your teacher — asked the same question that Vitruvius and Leonardo asked, but using modern mathematics. Let's find out what he discovered.”

STAGE 2 · Introduction (10 minutes)

Activity: Vocabulary Preview + Context Setting

Teacher introduces 10 key words. Students copy into notebooks.

Word	Meaning
approximation	a value that is close to, but not exactly, the correct answer
rational number	a number expressible as a fraction, e.g. 22/7
transcendental	a number (like π) that cannot be the solution of any algebraic equation
convergent	a fraction that gets progressively closer to a target value
proportion	a balanced relationship between parts
harmony	a pleasing, balanced arrangement
operator	a mathematical rule that transforms an input into an output
conjecture	a mathematical idea believed true but not yet proven
framework	a structured system of rules used to analyze problems
formalize	to write something precisely using strict logic or code

Context (2 min): “This paper goes back 3,650 years — to ancient Egypt. A scribe named Ahmes wrote that a circle with diameter 9 has roughly the same area as a square with side 8. That gives us $\pi \approx 256/81 \approx 3.1605$. The paper asks: does a modern mathematical system rediscover this ancient answer on its own? The answer is: partly.”

STAGE 3 · Presentation (20 minutes)**Activity: Guided Reading — The Vitruvian Approximation, pp. 1–2**

Students read the Abstract and Introduction silently (8 min). They underline any vocabulary words they recognize from Stage 2.

Comprehension check — individual (5 min): Students answer in writing.

5. What are the three honest goals the paper states?
6. What historical document does the paper reference, and how old is it?
7. What is the “Rhind approximation” of π , and how accurate is it?
8. What does the author mean by calling a conjecture “honestly marked sorry”?
9. In your own words: what is this paper really about?

Whole-class review (7 min): Teacher uses directed questions. Focus attention on the paper's intellectual honesty — the author states openly what is proven, what is not, and what remains a mystery. Connect to the video: Leonardo also worked at the edge of what was knowable.

STAGE 4 · Guided Practice (20 minutes)**Activity A — Vocabulary in Context (10 min)**

Students complete the worksheet using the 10 words from Stage 2.

10. The fraction $22/7$ is a well-known _____ for the value of π .
11. Mathematicians have proven that π is _____, meaning no polynomial equation with whole-number coefficients can produce it.
12. A _____ number can always be written as p divided by q , where both p and q are integers.
13. The continued-fraction expansion of π produces _____ like $333/106$ and $355/113$ that get closer and closer to the true value.
14. Vitruvius believed that _____ in architecture reflected the _____ of the human body.
15. A _____ is an idea that seems very likely to be true but has not yet been fully proven.
16. The G-cycle is an _____ chain that applies a series of transformations to circle data.
17. The research team used Lean 4 to _____ the theorems and make them verifiable by computer.
18. The dm^3 _____ provides a structured set of rules for modeling contact dynamics across many different systems.

Activity B — Grammar Focus: Passive Voice (10 min)

The paper uses the passive voice throughout (common in academic writing). Students rewrite these sentences in the passive.

19. Ahmes wrote the rule in the Rhind Papyrus around 1650 BCE.
20. Da Vinci solved the ancient puzzle of squaring the circle visually.
21. The team has not yet proven Conjecture 5.4.
22. Researchers formalized the theorems using the Lean 4 language.
23. Mathematicians classify π as a transcendental number.

STAGE 5 · Communicative Practice (20 minutes)**Activity: Discussion Cards — Groups of 3–4**

Each group picks one card. Discuss for 7 minutes, then present main ideas to class (2 min each).

CARD A Discovery vs. Recovery

The paper states that the G-cycle “packages the known ratio in operator language; it does not generate $8/9$ from a blank circle.” In other words, the framework finds the ancient answer — but only because the answer was already built into it. Discuss: What is the difference between discovering something new and recovering something already known? Does that difference matter in science? In your own learning?

CARD B Honest Sorry

The author marks one conjecture as “honestly sorry” — meaning he believes it is true but cannot yet prove it. He publishes the paper anyway. Discuss: Why might it be important for a scientist — or a student — to admit what they don't know? How does this connect to the dedication: “Neither of them read from a recipe book. Both of them ate the cake”?

CARD C Ancient Wisdom, Modern Math

The Rhind Papyrus is nearly 3,700 years old, yet a paper published in 2026 in Newark, NJ still refers to it. Discuss: What does this tell us about knowledge? Can old knowledge be more valuable than new knowledge? What examples from your own culture or language can you think of?

CARD D Newark to the World

This paper was written by your teacher, affiliated with Newark, NJ. It has been published as a preprint in an international mathematical series. Discuss: What does it mean to you that serious academic research can come from your community? Does knowing the author change how you read the paper? How?

STAGE 6 · Evaluation (15 minutes)**Activity: Exit Paragraph — Written Response**

Students write 8–10 sentences responding to ONE prompt.

Option A — Summary: Summarize The Vitruvian Approximation in your own words. What problem does it address? What does it find? What does it admit it cannot yet prove? Use at least 6 vocabulary words from today's lesson.

Option B — Personal Response: What idea from today's lesson — from the video, the paper, or the discussion — surprised you, moved you, or made you think differently? Connect it to your own life, background, or learning experience.

Option C — The Dedication: The paper opens: “Neither of them read from a recipe book. Both of them ate the cake. This is what education is, and what it is not.” Write a paragraph explaining what you think this means, and whether you agree.

Teacher collects paragraphs. Offer brief verbal feedback before closing the class.

STAGE 7 · Application (15 minutes)**Activity: “Your Own Approximation” — Pair Reflection + Share**

Students discuss in pairs, then write 3–5 sentences for each question.

24. Think of something important you learned by doing rather than by following instructions. What was it? How did it feel to figure it out yourself?
25. π cannot be written exactly — it goes on forever: 3.14159265... We use approximations (22/7, 355/113) because they are good enough for most purposes. Can you think of other areas of life — not just math — where a “good enough” answer is more useful than a perfect one that doesn't exist?
26. The paper was written over time, across continents, connecting ancient Egypt, Renaissance Italy, and modern Newark. Where do you come from, and what knowledge did you bring with you?

Volunteers share responses. Teacher closes: “Every approximation gets us closer to the truth. So does every lesson.”

Timing Summary

Stage	Activity	Time
1	Warm-Up — Video + Discussion	20 min
2	Introduction — Vocabulary + Context	10 min
3	Presentation — Guided Reading	20 min
4	Guided Practice — Vocabulary + Grammar	20 min
5	Communicative Practice — Discussion Cards	20 min
6	Evaluation — Exit Paragraph	15 min
7	Application — Your Own Approximation	15 min
TOTAL		120 min

Teacher Notes

- For stronger B2/C1 students: assign Sections 3–5 of the paper as optional extended reading, or ask them to identify additional examples of passive voice in the original text.
- For lower B2 students: allow the reading in pairs, simplify the vocabulary worksheet, and provide sentence starters for the discussion cards (“I think this matters because...” / “In my experience...” / “This reminds me of...”).
- The video can be paused and rewind as needed. Have a backup short clip in case of network issues.
- If the class is multilevel, use mixed-level pairs for Stage 5 (fluency) and same-level pairs for Stage 4 (accuracy).
- Recycle the dedication line as a closing phrase across the week — it builds continuity and rewards returning students.

Help

Pablo's trying to fund /h/House · 229 Ballantine Pkwy, Newark, NJ 07104

Lesson designed using the CAELA Framework for Effective Lesson Planning for Adult English Language Learners

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